

battery

Battery Management System

This page answers prerequisites information from **battery_discharge.csv** and **batter_flight.csv** about the battery model and properties to implement a Kalman based state of charge (SoC) state estimator

The implementation file is

[ekf_soc_estimator.py](#)

1. Battery Capacity

The battery capacity (mAh) or charge capacity is the total amount of electrical charge a battery can move through a circuit.

Identifying from the voltage-current discharge plot (Figure 1-1) the discharge current and the time duration of constant current until the sharp drop gives an estimate about the battery capacity.

$$Capacity = \frac{1.0 \text{ A} \times 6340 \text{ s}}{3600 \text{ s/hr}} = \mathbf{1.760 \text{ Ah}} = 1760 \text{ mAh}$$

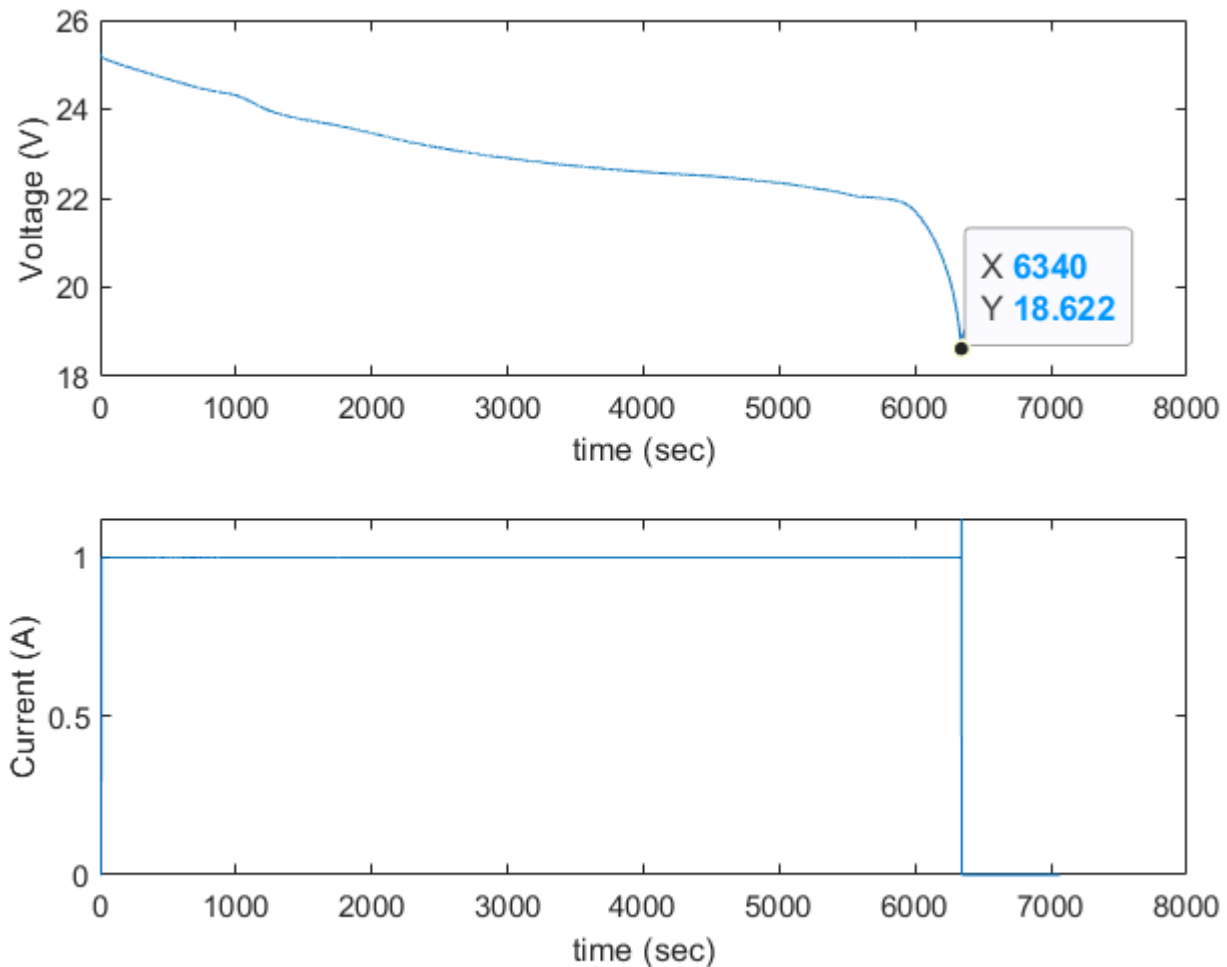


Figure 1-1: Voltage-current discharge

2. Battery topology

The battery topology that can be ascertained from the plots is that this is a typical lithium cell discharge profile connected in series to provide a maximum full discharge voltage of 25.22 V and a discharge cutoff voltage at around 18.6V. The typical LiPo cell is around 4.2 V fully charged.

So there are :

$$25.22 \text{ V} \div 4.2 \text{ V per cell} = \mathbf{6} \text{ cells in series}$$

The topology in terms of the voltage provision is 6 cells in series (6S).

In terms of capacity provision, the parallel topology is likely a single string of cells (1P) due to the fact that the capacity is 1800 mA. It is a 6S1P.

The charged battery at time $t < 9 \text{ sec}$ is 25.227V after which the current starts to flow (discharge) leading to immediate drop to 25.18V. Since the cells are serially connected, $V_{\text{cell}} = 4.197 \text{ V}$.

At $t = 6342$ sec, which is the moment the current drops to 0 A, is the transient response of the RC circuit, which gives us a clue about the τ (time constant of the dynamics of RC).

At this relaxation period, the τ is the 63.2% rise time from 18.648 (across the battery) to **19.82V**. In another word is τ is at voltage

$$V_{\tau} = 18.648 + 0.632 \times (19.814 - 18.648) = 19.39V$$

which is around $\tau = 6451 - 6340 = 111$ sec

If $R_0 = 25 \text{ m}\Omega$ and $R_1 = 12.5 \text{ m}\Omega$ then

$$C_1 = \frac{\tau}{R_1} = \frac{111 \text{ s}}{0.0125 \Omega} = 8880 \text{ F}$$

SoC Coulomb counting:

The coulomb counting provides a discrete approximation of the next state of charge and defined as:

$$SoC_k = SoC_{k-1} + \frac{I_k \cdot \Delta t}{Q_{nom} \cdot 3600} \quad | \quad I_k < 0 \text{ (discharge)}$$

where I_k and Q_{nom} are the current at instant k and the nominal battery capacity, respectively. The current I_k is negative of discharge.

Kirchhoff's Voltage Law (KVL):

This is cell voltage model that related the open source voltage (source) and the internal reactive components of the battery cell

$$V_{cell}(t) = OCV(SOC) + i(t) \cdot R_0 + v_1(t) \quad | \quad i < 0 \text{ (discharge)}$$

where the voltage is governed by a dynamic **first-order model (Thevenin Equivalent Model)**:

$$\frac{dv_1(t)}{dt} = -\frac{1}{R_1 C_1} v_1(t) + \frac{1}{C_1} i(t)$$

Because the topology does not change as there is no parallel configuration. The current is not divided, so the current flowing through the cells is the same.

from the [battery_flight.csv](#) data, the max cell voltage for the **4S1P** is 4.30 V.

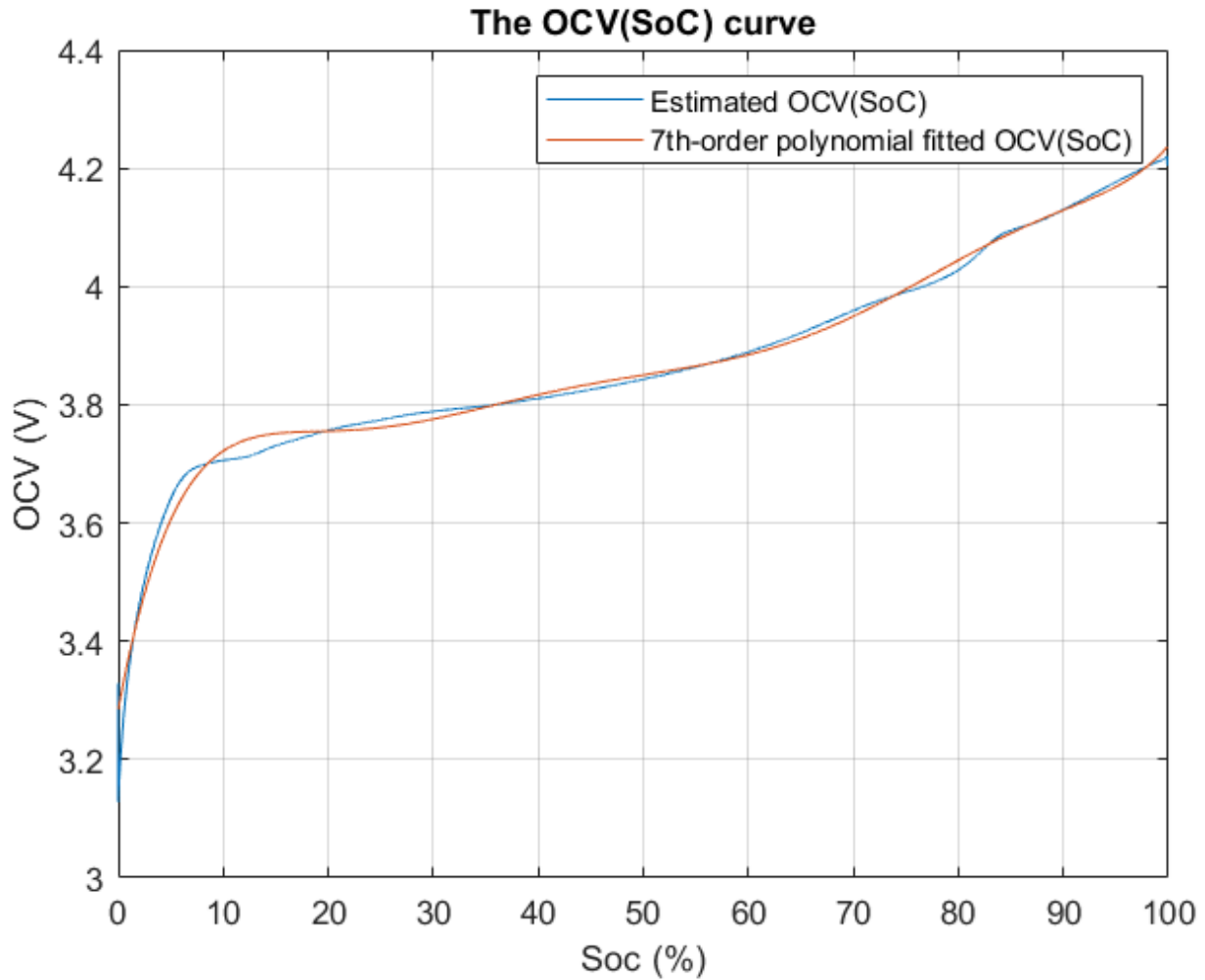


Figure 1-2: OCV function of SoC

Figure 1.2 shows the relationship between the OCV voltage computed from the Kirtchoff's law and the estimated SoC obtained from Coulomb Counting.

To find an estimate of the battery capacity from the current data, trapezoidal equation is used for discrete data points, the equation to sum up the area under your current-vs-time curve is:

$$Q_{\text{act}} = \frac{1}{3600} \sum_{k=2}^N \left(\frac{|I_k| + |I_{k-1}|}{2} \right) (t_k - t_{k-1})$$

The battery capacity of 4S1P with the energy/current draw provided, is **Q = 10000 mAh**

Because the current signal is noisy, the estimated would benefit from sliding window averaging or filter-based estimation.

3. Extended Kalman Filter

The estimator in [ekf_soc_estimator.py](#) uses a first-order Thevenin battery model and estimates two states for one cell: the RC voltage v_1 and state of charge $z = SoC$.

The measured pack voltage is divided by six before estimation; the plotted model voltage is converted back to pack voltage.

Parameters used

Parameter	Value	Meaning
Nominal capacity, Q_{nom}	10000 mAh	Coulomb-counting capacity
Series cells	4	Pack-to-cell voltage conversion
R_0	0.025 Ω	Cell ohmic resistance
R_1	0.0125 Ω	Cell polarization resistance
τ	111 s	RC time constant
$C_1 = \tau/R_1$	8880 F	Cell polarization capacitance
Initial SoC	0.999 (99.9%)	Initial state at the first sample
Initial polarization voltage	0 V	Initial state at the first sample
Initial covariance, P_0	$diag(0.01, 0.05)$	Variances for the state $[v_1, SoC]$
Process covariance, Q	$diag(10^{-4}, 10^{-7})$ per sample	Process/State-model uncertainty
Voltage measurement variance, R_v	$1.96 \times 10^{-6} \text{ V}^2$	Equivalent standard deviation: 10 mV per cell

The CSV current is positive for discharge, while the estimator defines positive current as charging.

My script uses therefore the negative sign to designate discharge

$$I_k = -I_{discharge,k}$$

With this convention, discharge current is negative and reduces SoC.

Regarding Q and P_0 , they are assumed because current and voltage measurement are not sources for characterizing the initial state error covariance and process covariance, since they require independent state references or explicit prior quantification

The measurement covariance or noise variance in this case was estimated from a detrended 60 seconds window portion of the voltage signal from "battery_flight.csv" data and estimated to be about 5.6 mV for the battery, or 1.4 mV per cell (4S1P).

The OCV function model used by the code is a high-order polynomial chosen so to handle the convex changes.

$$OCV(z) = \lambda^T z$$

where $z = [z^7, z^6, z^5, z^4, z^3, z^2, z^1, 1]^T$ and λ is the parameter vector.

EKF formulation

The state vector and its covariance are

$$x_k = \begin{bmatrix} v_{1,k} \\ z_k \end{bmatrix} \in R^2, \quad P_k = Cov(x_k) \in R^{2 \times 2}$$

For the elapsed time $\Delta t_k = t_k - t_{k-1}$, the zero-order-hold prediction uses the previous current sample (I_k remains constant during Δt_k).

Prediction:

Let $a_k = e^{-\Delta t_k / \tau}$.

The discrete state prediction (See [Appendix](#) for derivation) is

$$\begin{aligned} v_{1,k}^- &= a_k v_{1,k-1} + R_1(1 - a_k)I_{k-1}, \\ z_k^- &= z_{k-1} + \frac{\Delta t_k I_{k-1}}{3600 \times Q_{nom}} \quad ; z_k \in [0, 1], \\ F_k &= \begin{bmatrix} a_k & 0 \\ 0 & 1 \end{bmatrix}, \\ P_k^- &= F_k P_{k-1} F_k^T + Q \end{aligned}$$

The voltage measurement is the terminal cell voltage, not an OCV-corrected voltage. The predicted measurement and its Jacobian are

$$\begin{aligned} \hat{V}_k^- &= OCV(z_k^-) + v_{1,k}^- + R_0 I_k, \\ H_k &= \begin{bmatrix} 1 & \left. \frac{dOCV}{dz} \right|_{z_k^-} \end{bmatrix}, \\ \nu_k &= V_{cell,k} - \hat{V}_k^-, \\ S_k &= H_k P_k^- H_k^T + R_v, \\ K_k &= P_k^- H_k^T S_k^{-1} \end{aligned}$$

The gain K_k weights the voltage residual ν_k to correct both states.

The precomputed coefficients of the polynomial used to estimate $OCV(Soc)$ are:

Coefficient λ_i	P_7	P_6	P_5	P_4	P_3	P_2	P_1	P_0
Value	156.99	-599.64	924.67	-738.43	326.25	-78.49	9.61	3.28

Where P_i are the coefficient to the 7th degree polynomial. The Jacobian of which with respect the state is computed during the correction stage.

Correction:

$$x_k = x_k^- + K_k \nu_k$$

$$P_k = (I - K_k H_k) P_k^- (I - K_k H_k)^T + R_v K_k K_k^T$$

The SoC is clipped to $[0, 1]$ in the corrected step. For numerical stability, I use Joseph form for covariance update of P_k

The SoC in both the predicted or corrected step is clipped to $[0, 1]$. Also I zero in the implementation the SoC row and column of the covariance. The idea is to keep uncertainty consistent with projecting SoC back into its physical range $[0, 1]$, despite the fact that covariance is corrected at every sample.

Estimated voltage and SoC

The figure compares measured pack voltage with voltage reconstructed from the corrected EKF states, and shows the estimated SoC on the lower axis. It was generated from

`Telemetry/battery_discharge.csv`; the first sample uses the initialized state and subsequent samples perform prediction and voltage correction.

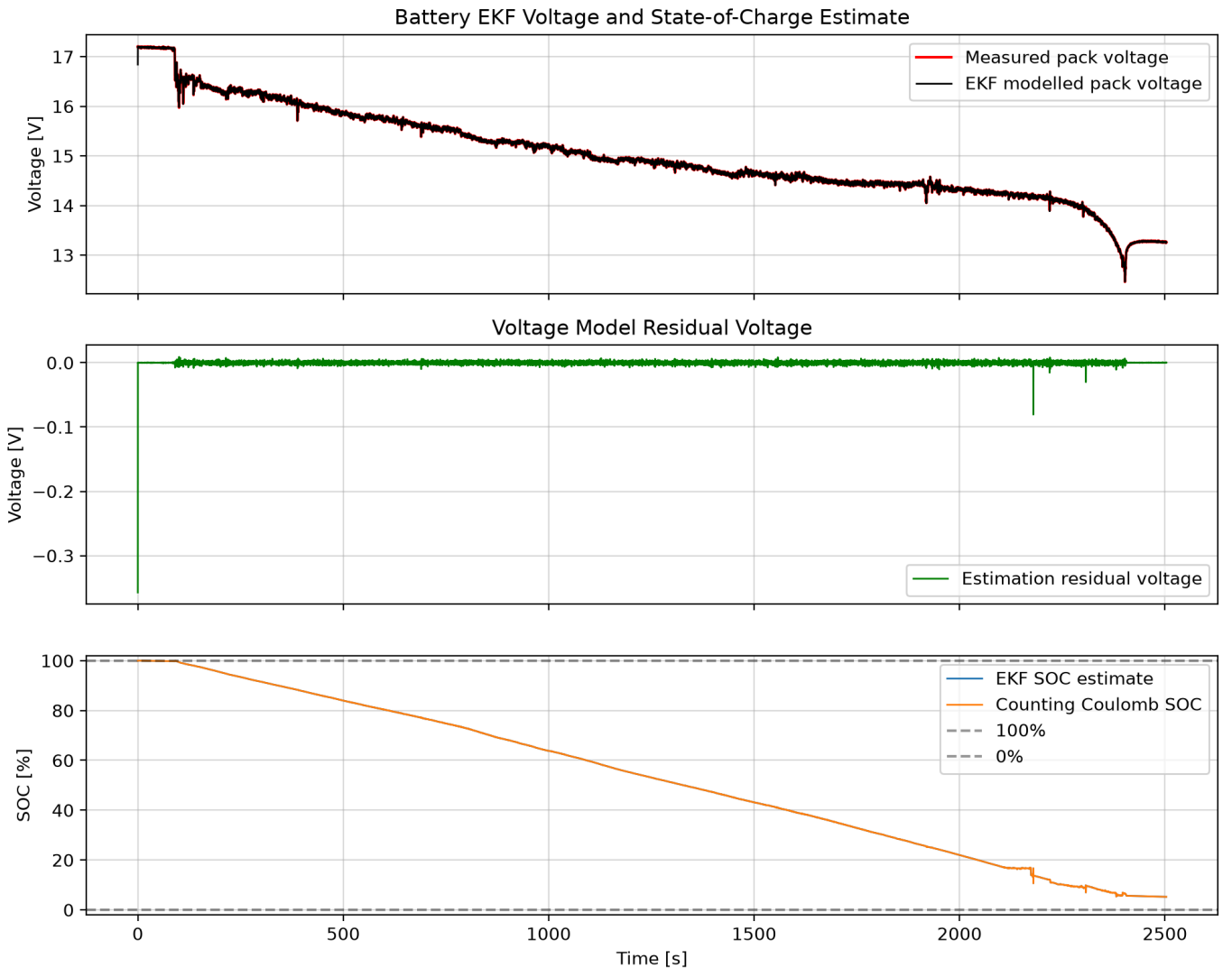


Figure 3-1: EKF measured and modelled pack voltage with estimated SoC

The upper plot is calculated as

$$\hat{V}_{pack,k} = 6 \times [OCV(z_k) + v_{1,k} + R_0 I_k]$$

It follows the overall discharge and relaxation profile, but the minimum modeled voltage is about 18.96 V while the measured minimum is 18.58 V; the low-voltage cutoff is therefore under-predicted by about 0.37 V. The lower trace falls from 99.90% to 30.88% over this data set. This is an EKF estimate, not a ground-truth SoC measurement; its accuracy depends on the capacity, OCV curve, RC parameters, current sign, and voltage data.

Appendix

Obtaining discrete solution of the first-order linear ODE with the current *ZOH* assumption

The continuous differential equation as stated above for the RC voltage

$$\frac{dV_1(t)}{dt} = -\frac{1}{\tau_1} V_1(t) + \frac{1}{C_1} I(t)$$

Where $\tau_1 = R_1 C_1$

Rearranging it into standard linear form gives:

$$\frac{dV_1(t)}{dt} + \frac{1}{\tau_1} V_1(t) = \frac{1}{C_1} I(t)$$

Multiplying the equation by an integrating factor $e^{\frac{t}{\tau_1}}$

$$e^{\frac{t}{\tau_1}} \frac{dV_1(t)}{dt} + \frac{1}{\tau_1} e^{\frac{t}{\tau_1}} V_1(t) = e^{\frac{t}{\tau_1}} \frac{1}{C_1} I(t)$$

The left side simplifies to a single derivative using the product rule:

$$\frac{d}{dt} \left[e^{\frac{t}{\tau_1}} V_1(t) \right] = e^{\frac{t}{\tau_1}} \frac{1}{C_1} I(t)$$

Integrating both sides in the duration of the sampling time

$$t_{k+1} - t_k = T_s$$

$$\int_{t_k}^{t_{k+1}} \frac{d}{dt} \left[e^{\frac{t}{\tau_1}} V_1(t) \right] dt = \int_{t_k}^{t_{k+1}} e^{\frac{t}{\tau_1}} \frac{1}{C_1} I(t) dt$$

Evaluating the left side directly yields:

$$\left[e^{\frac{t}{\tau_1}} V_1(t) \right]_{t_k}^{t_{k+1}} = e^{\frac{t_{k+1}}{\tau_1}} V_{1,k+1} - e^{\frac{t_k}{\tau_1}} V_{1,k}$$

The ZOH assumptions allows us to write:

$$\int_{t_k}^{t_{k+1}} e^{\frac{t}{\tau_1}} \frac{1}{C_1} I(t) dt = \frac{I_k}{C_1} \int_{t_k}^{t_{k+1}} e^{\frac{t}{\tau_1}} dt$$

Since the resistance R_1 :

$$R_1 = \frac{\tau_1}{C_1}$$

Then

$$\int_{t_k}^{t_{k+1}} e^{\frac{t}{\tau_1}} \frac{1}{C_1} I(t) dt = R_1 I_k \left(e^{\frac{t_{k+1}}{\tau_1}} - e^{\frac{t_k}{\tau_1}} \right)$$

Finally, with the left side and right side simplified:

$$e^{\frac{t_{k+1}}{\tau_1}} V_{1,k+1} - e^{\frac{t_k}{\tau_1}} V_{1,k} = R_1 I_k \left(e^{\frac{t_{k+1}}{\tau_1}} - e^{\frac{t_k}{\tau_1}} \right)$$

To solve for $V_{1,k+1}$, multiply every term in the equation by $e^{-\frac{t_{k+1}}{\tau_1}}$:

$$V_{1,k+1} - e^{\frac{t_k - t_{k+1}}{\tau_1}} V_{1,k} = R_1 I_k \left(1 - e^{\frac{t_k - t_{k+1}}{\tau_1}} \right)$$

Substituting $t_k - t_{k+1} = -T_s$ provides the discrete solution to the first-order ODE:

$$V_{1,k+1} = e^{-\frac{T_s}{\tau_1}} V_{1,k} + R_1 \left(1 - e^{-\frac{T_s}{\tau_1}} \right) I_k$$

References

1- Xiangdong Cui, Yanqi Huang, Xiaomei Wu, Kalman filter method with correction of open-circuit voltage curve for estimating SOC of lithium-ion batteries, Ain Shams Engineering Journal, Volume 16, Issue 12, 2025, 103822, ISSN 2090-4479, <https://doi.org/10.1016/j.asej.2025.103822>.